

# Advanced Querying and Analytics

## Introduction to SQL

### Window Functions



**Advanced Querying & Analytics**

Unlocking Insights from Data

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# Introduction to SQL Window Functions

A **window function performs** calculations across a set of table rows that are related to the current row. Unlike aggregate functions, window functions do not group rows into a single output, they add extra columns while preserving original data.

## Key Elements:

**PARTITION BY:** Breaks data into groups

**ORDER BY:** Defines calculation order within each partition

**OVER():** The "window" for the calculation

| Orders | Region | Amount | Total_By_Region |
|--------|--------|--------|-----------------|
| 101    | East   | 100.00 | 200.00          |
| 101    | East   | 60.00  | 200.00          |
| 101    | East   | 40.00  | 200.00          |
| 102    | West   | 50.00  | 50.00           |
| 103    | East   | 30.00  | 30.00           |

# Introduction to SQL Window Functions

## Window Frames in SQL:

- Default Frame:  
**RANGE BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW**
- Frame Types:
  - **RANGE**: Groups duplicate values as one unit
  - **ROWS**: Treats each row individually
- Key Options:
  - **CURRENT ROW**: The current row
  - **UNBOUNDED PRECEDING**: Start of the partition
  - **N PRECEDING**: N rows before the current row
  - **UNBOUNDED FOLLOWING**: End of the partition
  - **N FOLLOWING**: N rows after the current row